YINGAO (ELAINE) YAO

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https://elaineyao.github.io/

RESEARCH INTERESTS

My research interests are at the intersection of Software Engineering and Machine Learning (ML). I am particularly interested in developing software testing techniques (e.g., fuzzing) to improve the security, reliability, and correctness of ML systems.

EDUCATION

Cornell University *September 2024 - Present*
Ph.D. in Computer Science.
Advisor: Dr. Saikat Dutta.

University of British Columbia *September 2021 - December 2023*
MASc. Electrical and Computer Engineering.
Advisor: Dr. Karthik Pattabiraman.

University of Electronic Science and Technology of China *September 2017 - June 2021*
B.E. Electrical and Communication Engineering
GPA: 3.96/4.0

PUBLICATIONS

Automated Assertion Generation and Regression Testing for Machine Learning Notebooks *ASE 2026*
Yingao (Elaine) Yao, Vedant Nimje, Varun Viswanath, and Saikat Dutta.
IEEE/ACM International Conference on Automated Software Engineering.
🏅 **ACM SIGSOFT Distinguished Paper Award**

SwarmFuzz: Discovering GPS Spoofing Attacks in Drone Swarms *DSN 2023*
Yingao (Elaine) Yao, Pritam Dash, and Karthik Pattabiraman.
IEEE/IFIP International Conference on Dependable Systems and Networks (DSN).

Poster: May the Swarm Be With You: Sensor Spoofing Attacks Against Drone Swarms *CCS 2023*
Yingao (Elaine) Yao, Pritam Dash, and Karthik Pattabiraman.
ACM SIGSAC Conference on Computer and Communications Security (CCS).

PREPRINTS

ICS-Sniper: A Targeted Blackhole Attack on Encrypted ICS Traffic *arXiv 2023*
Gargi Mitra, Pritam Dash, Yingao Elaine Yao, Aastha Mehta, and Karthik Pattabiraman.
https://arxiv.org/abs/2312.06140.

HONORS AND AWARDS

ACM SIGSOFT Distinguished Paper Award *2026*
IEEE DSN Student Travel Grant Award *2023*
Faculty of Applied Science Graduate Award, University of British Columbia *2022*
Outstanding Graduate, Sichuan Province *2021*
Outstanding Winner, COMAP Interdisciplinary Contest in Modeling *2020*
Thanksgiving Scholarship for Modern Scientists *2020*

National Scholarship *2020*
National Second Prize in China Undergraduate Mathematical Contest in Modeling *2018*

WORKING EXPERIENCE

Security Researcher *Jan 2024 - Aug 2024*
Fortinet, Vancouver, Canada.

TEACHING EXPERIENCE

Teaching Assistant for CS 5150: Software Engineering *Spring 2025*
Cornell University, Ithaca, NY, USA

Teaching Assistant for CPEN 321: Software Engineering *Spring 2022, Fall 2022, Spring 2023*
University of British Columbia, Vancouver, Canada.

ACADEMIC SERVICE

Artifact Evaluation Program Committee *2024*
19th USENIX Symposium on Operating Systems Design and Implementation (OSDI).
European Conference on Object-Oriented Programming (ECOOP).
18th USENIX WOOT Conference on Offensive Technologies (WOOT).
ACM Conference on Computer and Communications Security (CCS).
USENIX Annual Technical Conference (USENIX ATC).
The 54th Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN).

Artifact Evaluation Program Committee *2023*
The 34th IEEE International Symposium on Software Reliability Engineering (ISSRE).

DEPARTMENT SERVICE

Department of Computer Science, Picnic Czar *Fall 2025,* **Spring 2026**
Cornell University, Ithaca, NY.

MENTORING

David Jojic (BS, University of Belgrade)
Arpan Chakraborty (BS, Adamas University)
Vedant Ramesh Nimje (BS, Veermata Jijabai Technological Institute, Mumbai)
Varun Viswanath (BS, Dwarkadas J Sanghvi College of Engineering)
Parshwa Nandu (BS, Dwarkadas J Sanghvi College of Engineering)

PRESENTATIONS

SwarmFuzz: Discovering GPS Spoofing Attacks in Drone Swarms *DSN 2023*

Poster: May the Swarm Be With You: Sensor Spoofing Attacks Against Drone Swarms. *CCS 2023*

OPEN-SOURCE CONTRIBUTIONS

SwarmFuzz: https://github.com/DependableSystemsLab/SwarmFuzz

NBTest: https://github.com/seal-research/NBTest